

SQL Analysis for IT Helpdesk Ticket Analytics

SQLite | SQL Queries

Project Objective

This notebook imports the cleaned IT helpdesk dataset into SQLite and performs SQL queries to answer common business questions.

```
from google.colab import files

uploaded = files.upload()
```

[Choose Files](#) | cleaned_ticket.csv
cleaned_ticket.csv(text/csv) - 14281690 bytes, last modified: 7/30/2026 - 100% done
 Saving cleaned_ticket.csv to cleaned_ticket.csv

```
import pandas as pd
import sqlite3
```

```
df = pd.read_csv("cleaned_ticket.csv")
```

Load Cleaned Dataset

```
df.head()
```

	Ticket_ID	Body	Department	Priority	Tags
0	0	Dear Customer Support Team,I am writing to rep...	Technical Support	high	['Account', 'Disruption', 'Outage', 'IT', 'Tec...
1	1	Dear Customer Support Team,I hope this message...	Returns and Exchanges	medium	['Product', 'Feature', 'Tech Support']
2	2	Dear Customer Support Team,I hope this message...	Billing and Payments	low	['Billing', 'Payment', 'Account', 'Documentati...

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
conn = sqlite3.connect("helpdesk.db")
```

```
df.to_sql(
    "Tickets",
    conn,
    if_exists="replace",
    index=False
)
```

```
29651
```

Query 1: Display the First Five Records

```
query = """
SELECT *
FROM Tickets
LIMIT 5;
"""

pd.read_sql(query, conn)
```

	Ticket_ID	Body	Department	Priority	Tags
0	0	Dear Customer Support Team,I am writing to rep...	Technical Support	high	['Account', 'Disruption', 'Outage', 'IT', 'Tec...
1	1	Dear Customer Support Team,I hope this message...	Returns and Exchanges	medium	['Product', 'Feature', 'Tech Support']
2	2	Dear Customer Support Team,I hope this message...	Billing and Payments	low	['Billing', 'Payment', 'Account', 'Documentati...

Query 2: Count Total Number of Tickets

```
query = """
SELECT COUNT(*) AS Total_Tickets
FROM Tickets;
"""
```

```
pd.read_sql(query, conn)
```

	Total_Tickets
0	29651

Query 3: Count Tickets by Department

```
query = """
SELECT Department,
COUNT(*) AS Total_Tickets

FROM Tickets

GROUP BY Department

ORDER BY Total_Tickets DESC;
"""
```

```
pd.read_sql(query, conn)
```

	Department	Total_Tickets
0	Technical Support	8617
1	Product Support	5539
2	Customer Service	4482
3	IT Support	3500
4	Billing and Payments	3017
5	Returns and Exchanges	1467
6	Service Outages and Maintenance	1157
7	Sales and Pre-Sales	885
8	Human Resources	568
9	General Inquiry	419

Query 4: Count Tickets by Priority

```
query = """
SELECT Priority,
COUNT(*) AS Total

FROM Tickets

GROUP BY Priority

ORDER BY Total DESC;
```

"""

```
pd.read_sql(query, conn)
```

	Priority	Total	
0	medium	12126	
1	high	11512	
2	low	6013	

Query 5: Display High-Priority Tickets

```
query = """
SELECT *
FROM Tickets
WHERE Priority='high';
"""
```

```
pd.read_sql(query, conn)
```

	Ticket_ID	Body	Department	Priority	Tags
0	0	Dear Customer Support Team,I am writing to rep...	Technical Support	high	['Account', 'Disruption', 'Outage', 'IT', 'Tec...
1	4	Dear Customer Support,I hope this message reac...	Technical Support	high	['Feature', 'Product', 'Documentation', 'Feedb...
2	5	Dear Customer Support Team,I am submitting a r...	Service Outages and Maintenance	high	['Outage', 'Disruption', 'Performance', 'IT', ...
3	9	Dear Customer Support Team,I hope this message...	Product Support	high	['Product', 'Service', 'Feedback', 'Sales']
4	12	Dear Customer Support Team,I am reaching out t...	Technical Support	high	['Feature', 'Account', 'Marketing', 'Performan...
...	...	...	...	...	...
11507	29646	Hello Customer Support Team,My name is name. I...	IT Support	high	['Technical Support', 'Urgent Issue', 'Service...
11508	29647	Dear Customer Support Team,We are currently en...	IT Support	high	['Service Disruption', 'IT Support', 'Urgent I...

Query 6: Display Low-Priority Tickets

```
query = """
SELECT *
FROM Tickets
WHERE Priority='low';
"""
```

```
pd.read_sql(query, conn)
```

Ticket_ID		Body	Department	Priority	Tags
0	2	Dear Customer Support Team,I hope this message...	Billing and Payments	low	['Billing', 'Payment', 'Account', 'Documentati...']
1	21	Dear Customer Support Team,I am reaching out t...	Technical Support	low	['Hardware', 'Tech Support', 'Driver', 'Audio']
2	25	Dear Customer Support Team,I hope this message...	Technical Support	low	['Documentation', 'Feature', 'IT', 'Tech Suppo...']
3	26	Dear Customer Support Team,I hope this message...	Customer Service	low	['Network', 'Disruption', 'Connectivity', 'Tec...']
4	27	Dear Customer Support Team,I hope this message...	Billing and Payments	low	['Billing', 'Payment', 'Dispute', 'Documentati...']
...	...	...	...	...	...
6008	29616	Dear Customer Support Team,I hope this message...	Technical Support	low	['Technical Support', 'Product Support', 'Soft...']
6009	29623	Hello Customer Support,I am reaching out to re...	IT Support	low	['Technical Support', 'IT Support', 'Product S...']

Query 7: Department-wise Priority Distribution

```
query = """
SELECT Department,
Priority,
COUNT(*) AS Total

FROM Tickets

GROUP BY Department, Priority

ORDER BY Department;
"""

pd.read_sql(query, conn)
```

	Department	Priority	Total
0	Billing and Payments	high	886
1	Billing and Payments	low	632
2	Billing and Payments	medium	1499
3	Customer Service	high	828
4	Customer Service	low	1464
5	Customer Service	medium	2190
6	General Inquiry	high	66
7	General Inquiry	low	246
8	General Inquiry	medium	107
9	Human Resources	high	59
10	Human Resources	low	271
11	Human Resources	medium	238
12	IT Support	high	1657
13	IT Support	low	347
14	IT Support	medium	1496
15	Product Support	high	1682
16	Product Support	low	1048
17	Product Support	medium	2809
18	Returns and Exchanges	high	328
19	Returns and Exchanges	low	546
20	Returns and Exchanges	medium	593
21	Sales and Pre-Sales	high	154
22	Sales and Pre-Sales	low	323
23	Sales and Pre-Sales	medium	408
24	Service Outages and Maintenance	high	818
25	Service Outages and Maintenance	low	143
26	Service Outages and Maintenance	medium	196
27	Technical Support	high	5034
28	Technical Support	low	993
29	Technical Support	medium	2590

Query 8: Department with the Highest Number of Tickets

```

query = """
SELECT Department,
COUNT(*) AS Tickets

FROM Tickets

GROUP BY Department

ORDER BY Tickets DESC

LIMIT 1;
"""

pd.read_sql(query, conn)

```

	Department	Tickets
0	Technical Support	8617

Query 9: Count Total Departments

```
query = """
SELECT COUNT(DISTINCT Department) AS Total_Departments
FROM Tickets;
"""

pd.read_sql(query, conn)
```

Total_Departments	
0	10



Query 10: Priority-wise Ticket Count

```
query = """
SELECT Priority,
COUNT(*) AS Total

FROM Tickets

GROUP BY Priority;
"""

pd.read_sql(query, conn)
```

	Priority	Total
0	high	11512
1	low	6013
2	medium	12126

